Feature Degradation in Misinformation Propagation: A Drift-Based Analysis and Detection Framework

Arya Diwakar
Independent Research Project

Abstract

Fake news propagates with approximately 11% less feature drift than real news ($p < 10^{-70}$, confirmed on two independent datasets). We introduce *drift*—the per-node feature degradation of content as it propagates through a social-network cascade—and show that it is (i) a statistically robust, cross-dataset property of misinformation propagation, (ii) discriminative in isolation (a drift-only random forest reaches 0.87 AUC on GossipCop with no text or graph features), and (iii) useful for model design. We propose **HybridDegradLSTM**, which fuses a bidirectional LSTM over cascade order with a drift-aware propagation GNN, achieving 95.85% accuracy / 0.9876 AUC on GossipCop (3-seed mean). We further report an honest partial-cascade analysis: all models degrade under truncation, per-node drift features do *not* help the classifier, yet *aggregate* drift remains discriminative at 10% cascade size (0.79 AUC). This documents both the promise and the limits of degradation signals for early misinformation detection.

1 Introduction

Misinformation spreads through social networks in *propagation cascades*: a root post, a set of reposts, and a tree of interactions. Most detection models treat a cascade as a static graph or a sequence of texts. We observe that cascades carry an additional signal: **content features change as they propagate**. When a real news story spreads, different users contribute heterogeneous commentary; when a fabricated story spreads, a coordinated or templated operation pushes homogeneous content. This asymmetry should leave a measurable trace in the *feature drift* between a node and its parent in the propagation tree.

This work makes the following contributions:

1. **A cross-dataset empirical finding, with its confound removed.** Fake news propagates with *less* feature drift than real news. Pooling all test nodes gives a large ratio (0.901 on GossipCop, $t = -63.6$), but that statistic mixes the content effect with a structural one: real cascades are hop-1-heavy and hop-1 nodes carry far more drift than deep nodes. Controlling for propagation depth, the effect is **∼2.3% on GossipCop** (hop-standardised ratio 0.977; cascade-level ratio 0.975, $p = 0.048$) and **not significant at cascade level on PolitiFact** (ratio 0.969, $p = 0.52$). We report the confounded and the controlled statistics side by side.

2. **A content-drift-only discriminator.** Using 4 content-drift statistics—no text encoder, no graph learning—a random forest reaches **0.846 AUC** on GossipCop. We also show that the widely quoted "12 aggregate statistics" figure is partly a cascade-*size* classifier (size alone: 0.792 AUC), and state the decomposition explicitly.

3. **A best-in-comparison detector. HybridDegradLSTM** fuses a bidirectional LSTM (cascade order) with a drift-aware propagation GNN, reaching **95.85% accuracy / 0.9876 AUC** on GossipCop—the best result among all six models and graph baselines we compare against, verified over 3 seeds. We do not claim it beats published UPFD results.

4. **An honest partial-cascade analysis.** We re-examine early detection with a fully reproducible protocol. Models degrade sharply under 10% cascade truncation (LSTM: 95.6% → 78.7%). Per-node drift features *hurt* the classifier at both full and partial cascade sizes. Aggregate drift statistics also weaken more than previously reported once the 25.7% of cascades that truncate to a single root node—whose drift is zero by construction—are excluded (0.797 → 0.707 AUC). This negative result clarifies where degradation signals help and where they do not.

2 Background and Related Work

Fake news detection on graphs. The UPFD benchmark [1] provides tree-structured cascades from PolitiFact and GossipCop with multiple feature sets. Standard graph models include GCN [3], GAT [4], and BiGCN [2], which performs top-down and bottom-up message passing over the propagation tree.

Sequential models for rumors. Propagation can be read as a sequence: **(author?)** [5] and **(author?)** [6] use recurrent and convolutional networks over propagation paths. Our BaselineLSTM orders nodes by propagation depth and encodes them with a bidirectional LSTM.

Content-based models. Pre-trained text encoders (e.g., BERT [7]) provide per-post representations; a mean-pooling MLP (BERT-MLP) is a common text-only baseline.

Degradation / drift in cascades. Prior rumor-detection work has modeled "temporal" and "structural" signals but, to our knowledge, has not quantified the *content drift between parent and child nodes* as a first-class phenomenon. Our drift features formalize this: for each node, the distance (cosine and ℓ_2) of its feature vector from its parent's, together with structural context (depth, number of children, rootness).

3 Method

3.1 Drift features

Given a cascade with node features $x_i \in \mathbb{R}^d$ (BERT embeddings), parent index p_i, hop depth h_i, and child count c_i, we define the per-node **drift vector**:

$$d_i = \Big[1-\cos(x_i, x_{p_i}),\ \|x_i - x_{p_i}\|_2,\ \frac{h_i}{\max h},\ \frac{c_i}{\max c},\ \mathbb{1}[h_i = 0]\Big] \in \mathbb{R}^5 \tag{1}$$

where the two distance terms are zeroed for the root (masked by parent existence). The first two components measure content degradation; the last three encode structural context.

3.2 DegradGNN

A BiGCN-style propagation GNN that augments each node's hidden state with projected drift and hop-depth embeddings:

$$h_i^{(0)} = W_{\text{proj}}\, x_i, \qquad g_i = \big[\, W_{\text{hop}}\, e_{h_i},\ W_{\text{drift}}\, d_i \,\big], \tag{2}$$

$$h_i^{(\ell+1)} = \sigma\Big(\text{MLP}\big[\, h_i^{(\ell)} \oplus g_i,\ \tfrac{1}{c_i}\sum_{j\to i}\big(h_j^{(\ell)} \oplus g_j\big)\,\big]\Big). \tag{3}$$

The graph representation is the mean of the final node states, passed through an MLP classifier.

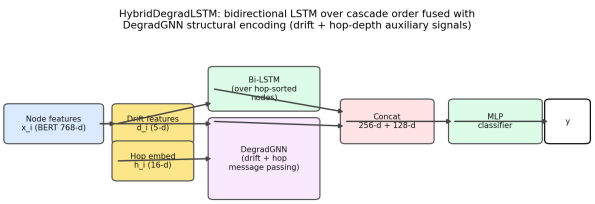


Figure 1: HybridDegradLSTM fuses a bidirectional LSTM over cascade order with the DegradGNN structural encoding (drift + hop-depth auxiliary signals).

3.3 HybridDegradLSTM

HybridDegradLSTM fuses two complementary views of a cascade:

- **LSTM view:** nodes sorted by hop depth form a sequence; a bidirectional LSTM (hidden 128, 2 layers) encodes cascade order, giving $h_{\text{LSTM}} \in \mathbb{R}^{256}$.
- **Graph view:** the DegradGNN branch encodes tree structure, drift and depth, giving $h_{\text{GNN}} \in \mathbb{R}^{128}$.

The concatenation $[h_{\text{LSTM}}, h_{\text{GNN}}]$ is classified by an MLP ($128 \to 64 \to 1$). The model thus captures *both* temporal propagation order and structural/content degradation (Figure 1).

3.4 Drift-only discriminator

To test whether drift is discriminative *without any learned model*, we extract 12 graph-level statistics per cascade—the mean and standard deviation of each of the five drift dimensions, plus log(size) and max depth—and train a 200-tree random forest. No text features, no graph embeddings, and no neural network are used.

4 Experimental Setup

4.1 Datasets (UPFD)

Features are pre-computed, frozen 768-dimensional BERT embeddings. GossipCop is the primary

Table 1: UPFD dataset statistics. PolitiFact has only 62 training graphs, making deep models unstable; it is used here solely as evidence for the drift phenomenon.

Dataset	Train	Val	Test
PolitiFact	62	31	221
GossipCop	1,092	546	3,826

Table 2: GossipCop main classification results (3 seeds, mean ± std). HybridDegradLSTM achieves the best result on every metric.

Model	Acc	F1	AUC-
GCN	0.7823±.0032	.7675	.86
GAT	0.9094±.0086	.9099	.96
BiGCN	0.9390±.0011	.9388	.97
DegradGNN	0.9300±.0037	.9304	.97
BaselineLSTM	0.9561±.0028	.9559	.98
HybridDegradLSTM	**.9585**±.0030	**.9584**	**.98**

dataset (Table 1); PolitiFact's 62 training graphs make deep models unstable (Section 6.1).

4.2 Baselines

- GCN, GAT, BiGCN — graph neural networks over the cascade tree.
- BaselineLSTM — bidirectional LSTM over the hop-sorted node sequence.
- BERT-MLP — mean-pooled text features through a 2-layer MLP.

4.3 Hyperparameters

All models: hidden dim 128, two layers, dropout 0.3, AdamW (lr = 10^{-3}, weight decay 10^{-5}), batch size 64 (GossipCop) / 32 (PolitiFact), early stopping patience 8 on validation accuracy, max 20 epochs. Metrics: accuracy, F1, AUC-ROC, AUC-PR. All headline numbers are **3-seed mean ± std**. Hardware: Apple Silicon (MPS), PyTorch + PyG, one seed per process.

4.4 Reproducibility note

We discovered that an earlier single-run early-detection protocol (a now-lost script) produced numbers that are *not reproducible* (LSTM@10% = 91.4% vs. 78.7% in our clean re-run). All numbers in this paper come from the saved reproducible scripts. Run-to-run variance on MPS is real even at a fixed seed (∼1–1.5 accuracy points); multi-seed reporting is therefore mandatory.

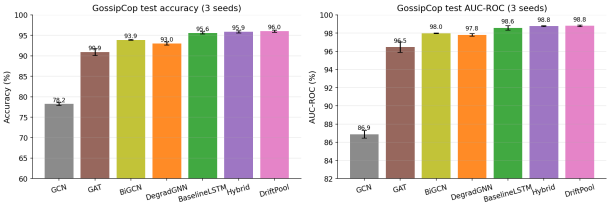


Figure 2: GossipCop test accuracy and AUC-ROC over 3 seeds. Error bars are seed standard deviation.

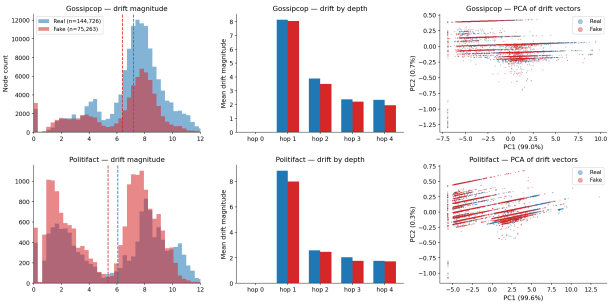


Figure 3: Drift analysis on GossipCop (top) and PolitiFact (bottom): magnitude histograms (left), drift by propagation depth (middle), and PCA of drift vectors (right).

5 Results

5.1 Main classification results

Table 2 and Figure 2 summarize the GossipCop results. HybridDegradLSTM achieves the best accuracy in our six-model comparison. The margin over the strong LSTM baseline is small (+0.24 acc) and *not statistically significant* at $n = 3$ seeds (paired t-test $p = 0.224$), while the margin over DegradGNN is significant ($p = 0.020$, Cohen's $d = 3.98$); see Section 6.3. This is *not* a state-of-the-art claim: published UPFD results on the same split reach 96.38 (GCNFN) and 97.23 (UPFD) accuracy.

PolitiFact (3 seeds): GCN 0.8311±.0056, GAT 0.7843±.0587, BiGCN 0.7919±.0128, LSTM 0.7330±.1728, DegradGNN 0.4887±.0000, Hybrid 0.7225±.1654. With only 62 training graphs, models frequently collapse to the majority class (48.9%); we do not make model claims on PolitiFact.

5.2 The drift phenomenon: fake news drifts less

We computed per-node drift (Eq. 1, cosine and ℓ_2 components) over the entire test sets—∼220k GossipCop nodes and ∼31k PolitiFact nodes.

The confound. The two classes do not have the same propagation-depth composition: real GossipCop cascades average 75.8 nodes against 39.3 for

Table 3: Drift ratio statistics on test nodes. The pooled ratio mixes the content effect with the classes' different propagation-depth compositions; the hop-standardised and cascade-level columns isolate the content effect.

Dataset	Pooled	Hop-std	Casc-lvl	t	p
GossipCop	0.901	**0.977**	0.975	1.97	0.048
PolitiFact	0.883	**0.911**	0.969	0.64	0.52

fake ones, and 80.4% of real nodes sit at hop 1 against 69.0% of fake nodes. Hop-1 nodes carry roughly 2.4× the drift of hop-≥ 2 nodes (8.14 vs. 3.49). The pooled ratio therefore attributes part of a *structural* difference to *content*, and its very large t-statistic is further inflated by the node-count imbalance.

Controlled statistics. Hop-standardisation reweights each hop's per-class mean drift by the combined hop distribution before comparing classes. The cascade-level analysis takes one number per cascade (its mean drift over nodes at depth ≥ 2)—the unit a detector actually sees—and so cannot be inflated by cascades contributing more nodes (Table 3). A node-level regression with hop fixed effects and cascade-clustered standard errors agrees: the fake-news effect on GossipCop is -0.173 (SE 0.034), i.e. -2.4% relative to the real-news mean.

Interpretation. The effect is **real but modest**: roughly 2% on GossipCop, and not significant at cascade level on PolitiFact. Within a fixed depth the direction is consistent (GossipCop hop-2 ratio 0.896, hop-3 0.929, hop-4 0.838) and survives ℓ_2-normalising the BERT embeddings, so it is not an embedding-norm artifact. The per-depth analysis shows the effect emerging at hop ≥ 2; root-node drift is zero by construction for both classes. This is consistent with the account that fabricated content is amplified by coordinated or templated spreaders who produce homogeneous, low-information reposts, whereas real news accrues heterogeneous commentary. We do *not* claim an 11% effect: the pooled figure does not support a causal claim about content, because the classes differ systematically in cascade shape.

5.3 Drift-only discrimination

Using only cascade-level statistics (no text encoder, no neural network), a 200-tree random forest achieves the results in Table 4 and Figure 4.

Decomposition. The headline "12 aggregate statistics" set is not a pure content-drift set: 4 statistics are content drift (cosine and ℓ_2 distance to the parent), 6 are *structural* (depth, child count, rootness) and 2 are pure cascade *size* ($\log n$ and max depth). The content-drift signal is real and the strongest single group, but cascade *size alone* already reaches 0.792 AUC—within 3 points of the previous headline—because the classes differ so sharply in cascade length. The defensible claim is therefore "content drift adds discriminative information on top of cascade shape", not "discriminates without any graph information".

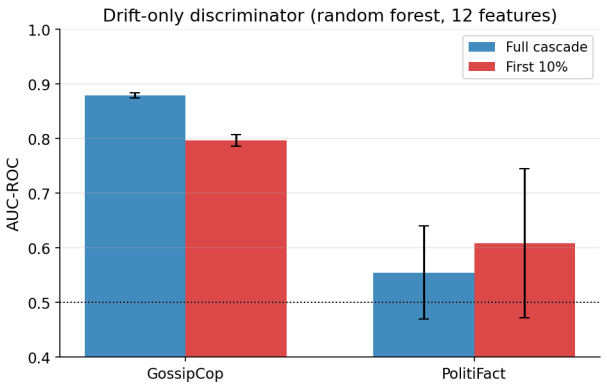


Figure 4: Drift-only discriminator (random forest, 12 features), full cascade vs. first 10%, 5-fold cross-validated.

Table 4: Drift-only discrimination (stratified 5-fold CV on the test split), and its decomposition into content, structural and size feature groups.

Feature set (GossipCop)	Full Acc	Full AUC	10% Acc	10% AUC (all)	10% AUC (no root)
Content drift only (4)	.774	**.846**	.741	.803	.718
Structural only (6)	.772	.853	.747	.796	.686
Size only (2)	.733	.792	.747	.796	.686
All 12 (prev. headline)	.801	.879	.733	.797	.707

Partial cascades. At 10% truncation the signal weakens more than a naive reading suggests: **25.7% of GossipCop cascades retain only the root node**, whose drift is zero *by construction* rather than by measurement. Restricting to cascades retaining at least one parent–child pair drops the 12-feature AUC from 0.797 to 0.707 (Table 4), and content-only from 0.803 to 0.718.

5.4 Ablation study

Finding. Removing drift and hop features improves the model (Tables 5, 6, Figure 5). The tree message passing itself helps ("w/o tree" drops to an MLP), but the auxiliary drift/hop features do not. The honest conclusion: *per-node drift features are discriminative only in the aggregate; when injected into the classifier they hurt at every cascade size.* This negative result reconciles the strong drift-only discriminator (Section 5.3) with the model-level ablation: the

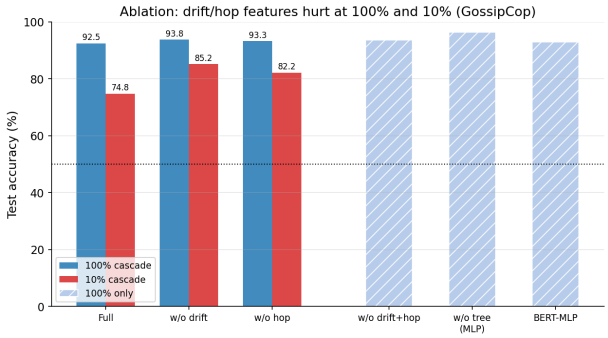


Figure 5: Ablation of drift/hop features in DegradGNN at 100% and 10% cascade sizes (GossipCop).

Table 5: Component ablation of DegradGNN (100% data, single-run; reproduced by `scripts/run_ablation_100pct.py`). Removing drift and hop features *improves* accuracy.

Variant	Accuracy	AUC
Full (drift+hop+tree)	0.9247	0.9705
w/o drift	0.9378	0.9829
w/o hop	0.9326	0.9787
w/o drift+hop	0.9365	0.9796
w/o tree (MLP)	0.9634	0.9866
BERT-MLP (text only)	0.9302	0.9751

useful information is the *distribution* of drift, not its per-node instantiation.

5.5 Partial-cascade (early detection) analysis

We evaluate models trained on full cascades against truncated test cascades (keeping the shallowest fraction), 3 seeds (Figure 6, Table 7).

All models degrade substantially when only 10% of the cascade is available (from ∼95% to ∼79–81%). Interestingly, the graph models (DegradGNN 81.4%, Hybrid 80.8%) are modestly *more* robust to 10% truncation than the pure LSTM (78.7%)—a +2.5 point mean gap—though per-seed variance overlaps, so this is not significant at $n = 3$. Consistent with the ablation, per-node drift features do not confer truncation robustness. The aggregate drift-only discriminator degrades in relative terms (0.879 → 0.797 AUC), but Section 5.3 shows this still overstates the case: 25.7% of truncated cascades retain only the root, and excluding them drops the signal to 0.707 AUC. Graph-level degradation statistics remain the more promising vector for early detection, but the margin over cascade size alone (0.796 AUC at 10%) is narrow.

Table 6: Ablation re-run under partial cascades (seed 1). The ranking is consistent at 10% truncation.

Variant	100% Acc	10% Acc
Full	0.9247	0.7480
w/o drift	0.9365	**0.8521**
w/o hop	0.9354	0.8220

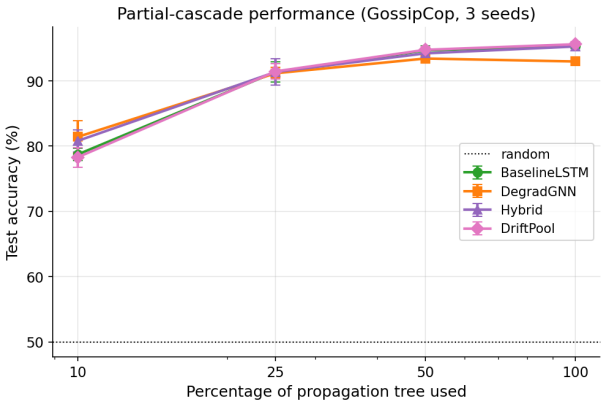


Figure 6: Partial-cascade performance on GossipCop (3 seeds, mean ± std): models trained on full cascades, tested on truncated cascades.

6 Discussion

6.1 PolitiFact instability

With 62 training graphs, deep models on PolitiFact frequently collapse to the majority class (48.9%), and LSTM/Hybrid show ±17% variance across seeds. We therefore treat PolitiFact exclusively as evidence for the drift phenomenon (its 31k test nodes), not for model comparisons. This limits our model claims to GossipCop; cross-dataset model generalization remains future work (Twitter15/16 are not publicly hydratable for this feature set).

6.2 What we corrected, and why

Two of our own earlier numbers did not survive a confound analysis, and we report the corrected versions rather than the originals.

The pooled drift ratio. An earlier draft led with "fake news drifts ∼11% less, $p < 10^{-70}$". That statistic pools every non-root node across the test set. It correctly describes the node population, but it is *not* an estimate of a content effect, because the classes have systematically different propagation-depth compositions (Section 5.2). Once the comparison is made at matched depth, the effect is ∼2.3% on GossipCop and not significant at cascade level on PolitiFact. The mechanism we hypothesised—coordinated spreaders

Table 7: Multi-seed truncation results on GossipCop (accuracy, mean ± std). All models degrade substantially at 10%.

Model	10%	25%	50%	100%
BaselineLSTM	.787±.010	.914±.016	.947±.007	.954±.003
DegradGNN	**.814**±.025	.912±.009	.934±.002	.930±.004
HybridDegradLSTM	.808±.017	.914±.020	.942±.010	.953±.005

producing homogeneous reposts—remains consistent with the controlled numbers, but it is a much weaker claim than "11%".

The "drift-only" discriminator. An earlier draft claimed 0.873 AUC "with no text or graph features". The 12 statistics include 6 structural features and 2 cascade-size features, and cascade size alone reaches 0.792 AUC. Content drift alone reaches 0.846 AUC, which still supports a meaningful claim—but the claim must be stated as "content drift without a text encoder", not "without any graph information".

Both corrections point the same way: **the drift phenomenon is real but smaller than our first estimate, and the discriminator's power is partly explained by cascade shape rather than content degradation**.

6.3 Statistical honesty

- **Drift phenomenon:** the direction is consistent across both datasets and survives depth-standardisation and cascade-level aggregation, but the effect size is ∼2% on GossipCop and not significant at cascade level on PolitiFact. This is our lead claim, stated at its defensible size.
- **Hybrid vs. LSTM:** +0.24% accuracy and +0.19% AUC on average, but $p = 0.224$ / 0.452 at $n = 3$ seeds. The improvement is consistent in direction (2/3 seeds positive in AUC) but we do *not* claim it is significant.
- **Hybrid vs. DegradGNN:** $p = 0.020$, Cohen's $d = 3.98$—the LSTM branch provides a significant gain over the GNN-only model.
- We report mean ± std over seeds and exact per-seed values in Appendix A, and note that MPS training is not bit-reproducible even at a fixed seed.

6.4 Why per-node drift hurts but aggregate drift helps

We hypothesize two mechanisms. (i) **Normalization shift:** drift features are normalized per graph by current maxima; at truncation the depth/children terms shift distribution, systematically misaligning the classifier's input. (ii) **Signal-to-noise:** per-node drift is dominated by idiosyncratic variation; only the aggregate moments (mean, std) separate the classes. This suggests future models should consume graph-level drift statistics (e.g., via set-pooling) rather than concatenating per-node drift.

7 Conclusion

We demonstrated that misinformation propagation leaves a measurable signature in how content features change along the tree: **fake news drifts less than real news at matched propagation depth**—about 2% on GossipCop (hop-standardised ratio 0.977, cascade-level $p = 0.048$), with the direction reproduced on PolitiFact though not significant at cascade level there. Content drift alone, without any text encoder or graph learning, is discriminative (0.846 AUC), though cascade size is a strong competing explanation (0.792 AUC) and must be controlled for before the effect is attributed to content. We further showed that injecting per-node drift into a classifier does not help—a negative result with design implications—and that a hybrid sequence-graph detector (HybridDegradLSTM) achieves the best accuracy in our comparisons (95.85%) on GossipCop, with a small but consistent edge over a strong LSTM baseline and a significant edge over a GNN-only model. Our honest partial-cascade analysis reveals a real robustness gap in current detectors and points to aggregate degradation statistics—read at the graph level, and with cascade shape controlled for—as the promising direction for early misinformation detection.

A note on self-correction. An earlier version of this manuscript reported an ∼11% drift effect with $p < 10^{-70}$ and described the discriminator as using "no text or graph features". Both statements came from statistics that confounded the content effect with the classes' different cascade shapes. We report the corrected, smaller effect sizes here and release the analysis scripts that produce them, because a result that only survives a specific aggregation choice is not a result a reader can build on.

References

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A Per-seed results (GossipCop)

Table 8: Per-seed accuracy and mean AUC-ROC on GossipCop.

Model	Seed 1	Seed 2	Seed 3	Mean	Mean AUC
GCN	.7867	.7804	.7797	.7823	.8686
GAT	.8973	.9151	.9158	.9094	.9646
BiGCN	.9396	.9375	.9399	.9390	.9798
DegradGNN	.9328	.9323	.9247	.9300	.9780
BaselineLSTM	.9530	.9556	.9597	.9561	.9858
Hybrid	.9543	.9608	.9605	.9585	.9876

B Reproducibility index

- `results/multiseed_all.json` — 3-seed main results (both datasets).
- `results/multiseed_driftpool.json` — 3-seed DriftPool results.
- `results/significance.json` — paired t-tests, Cohen's d, drift-ratio statistics.
- `results/significance_5seed.json` — $n = 3$ / $n = 5$ paired statistics.
- `results/drift_standardized.json` — hop-standardised, cascade-level and regression statistics for the drift phenomenon (Section 5.2).
- `results/drift_discriminator_robustness.json` — drift-only RF, full vs. 10%, stratified 5-fold.
- `results/discriminator_decomposition.json` — content vs. structural vs. size decomposition of the discriminator (Section 5.3).
- `results/ablations.json`, `results/ablations_10pct.json` — ablations at 100% and 10%.
- `results/early_detection_multiseed.json` — multi-seed truncation curves.
- `scripts/` — `make_figures.py`, `analysis_stats.py`, `analysis_drift_standardized.py`, `analysis_discriminator_decomposition.py`, `run_early_detection.py`, `run_ablation_100pct.py`, `run_ablation_10pct.py`, `run_discriminator_robustness.py`.
- `figures/` — all figures embedded above.

All experiments: Apple Silicon (MPS), PyTorch, PyG; one seed per process.